Practice workbook · Angle between two planes using a right-angled triangle

Finding the angle between two planes, where one is usually the base, by taking perpendiculars to their common line and working in a right-angled triangle.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-16

1. Question 9 · part whole · 6 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

9

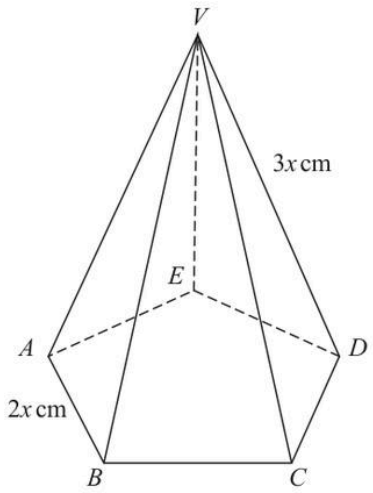


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid with vertex V and base $ABCDE$ which is a regular pentagon.

$$AB = BC = CD = DE = EA = 2x \text{ cm}$$
$$VA = VB = VC = VD = VE = 3x \text{ cm}$$

Find, in degrees to one decimal place, the size of the angle between the plane VBC and the base $ABCDE$

(6)

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1. Question 9 · part whole · 6 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 6 marks)

1. Question 9 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 3 · part c · 3 marks

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Further practice · Attempt part c only.

3

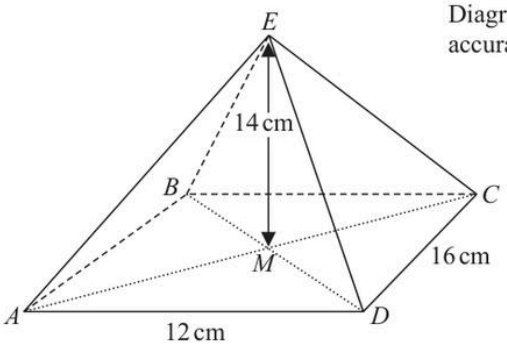


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the right pyramid $ABCDE$. The base, $ABCD$, of the pyramid is a horizontal rectangle with $AD = 12$ cm and $CD = 16$ cm. The height ME of the pyramid is 14 cm where M is the point of intersection of the diagonals of the base.

The sloping edges, EA, EB, EC and ED of the pyramid are all of equal length.

(a) Calculate, to 3 significant figures, the length of a sloping edge. **(3)**

Calculate, in degrees to one decimal place, the size of

(b) the angle between AE and the base, **(3)**

(c) the angle between the plane AED and the base. **(3)**

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2. Question 3 · part c · 3 marks

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Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 9 marks)

2. Question 3 · part c · 3 marks · Working and answer

Working and answer

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3. Question 8 · part c · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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Diagram **NOT** accurately drawn

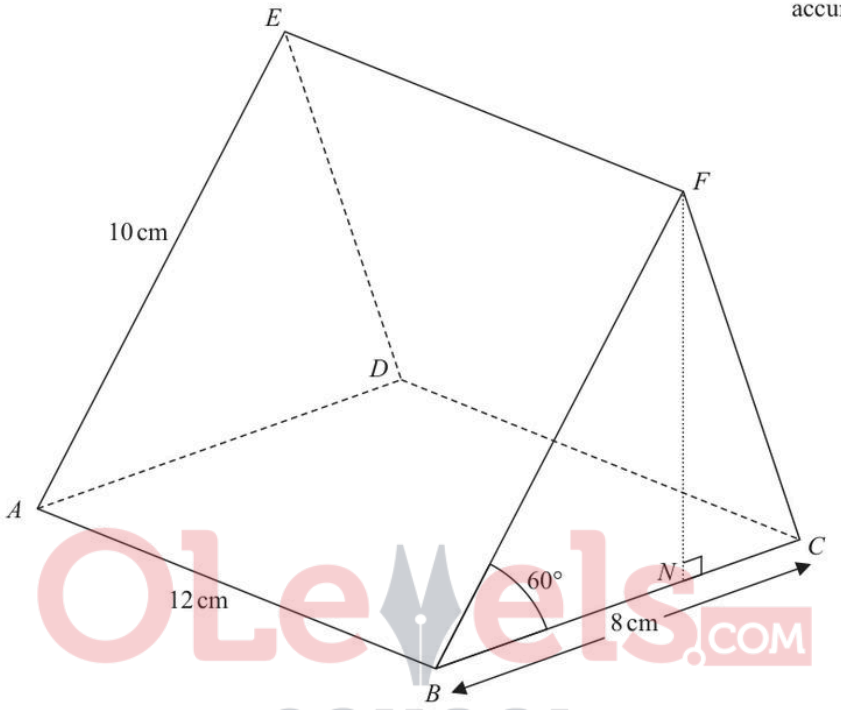


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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3. Question 8 · part c · 2 marks · Working and answer

Working and answer

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4. Question 9 · part c · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

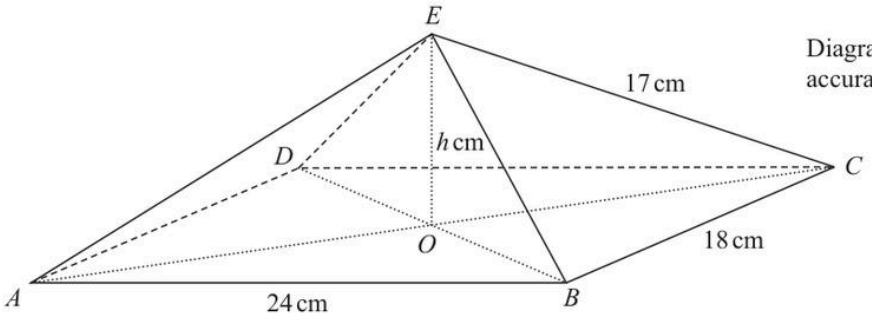


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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4. Question 9 · part c · 2 marks · Working and answer

Working and answer

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5. Question 9 · part d · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

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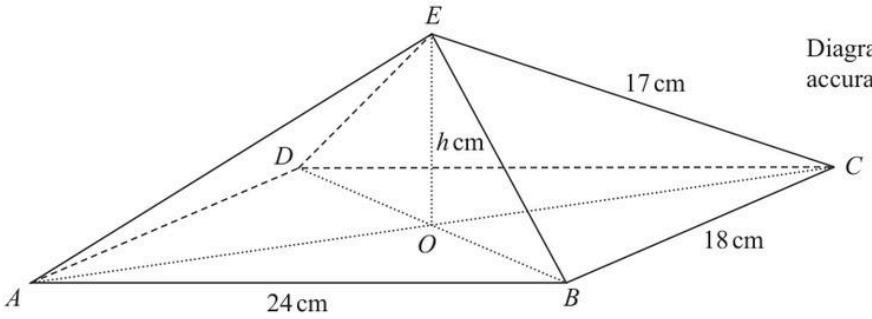


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Figure 1

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The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

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The vertex E of the pyramid is vertically above O such that

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The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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5. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

Question 9 continued

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5. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

Question 9 continued

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5. Question 9 · part d · 4 marks

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(Total for Question 9 is 10 marks)

5. Question 9 · part d · 4 marks · Working and answer

Working and answer

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6. Question 6 · part whole · 6 marks

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Further practice · Attempt part whole only.

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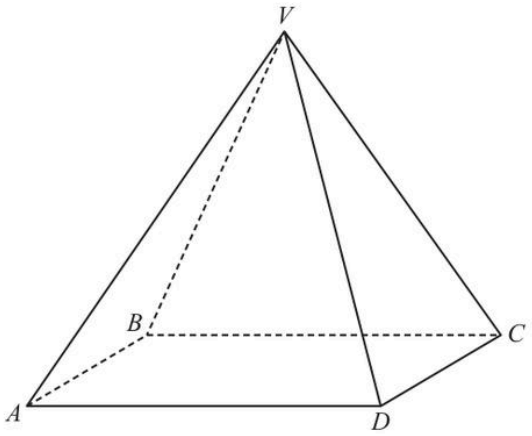


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a right pyramid $VABCD$ with vertex V and rectangular base $ABCD$

$$AD = VC = 3AB$$

Find in degrees, to one decimal place, the size of the angle between the plane CVD and the base $ABCD$

(6)

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6. Question 6 · part whole · 6 marks

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Further practice · Attempt part whole only.

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Question 6 continued

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6. Question 6 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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6. Question 6 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

6. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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7. Question 11 · part d · 3 marks

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Further practice · Attempt part d only.

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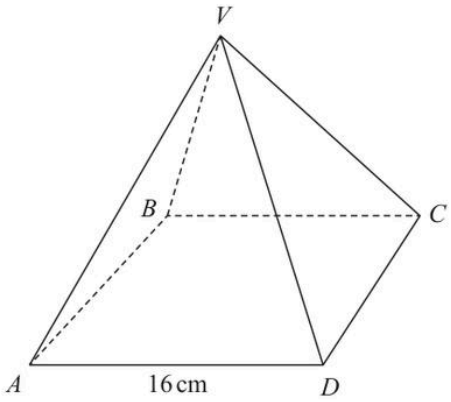


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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7. Question 11 · part d · 3 marks · Working and answer

Working and answer

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8. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

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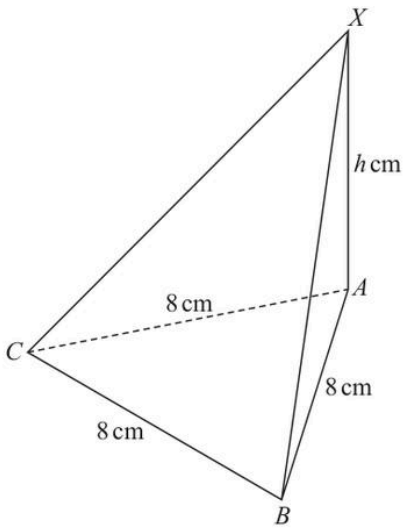


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Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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8. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

Question 4 continued

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8. Question 4 · part c · 3 marks

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Further practice · Attempt part c only.

Question 4 continued

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8. Question 4 · part c · 3 marks

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8. Question 4 · part c · 3 marks · Working and answer

Working and answer

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9. Question 6 · part whole · 6 marks

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Further practice · Attempt part whole only.

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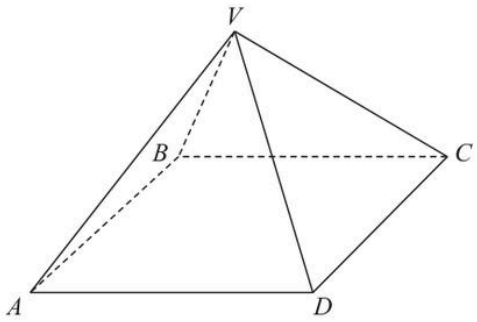


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid $VABCD$ with vertex V and square base $ABCD$.

Each of the edges of the pyramid has the same length.

Find the size, in degrees to one decimal place, of the angle between the plane CVD and the base $ABCD$.

(6)

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9. Question 6 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

9. Question 6 · part whole · 6 marks · Working and answer

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10. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

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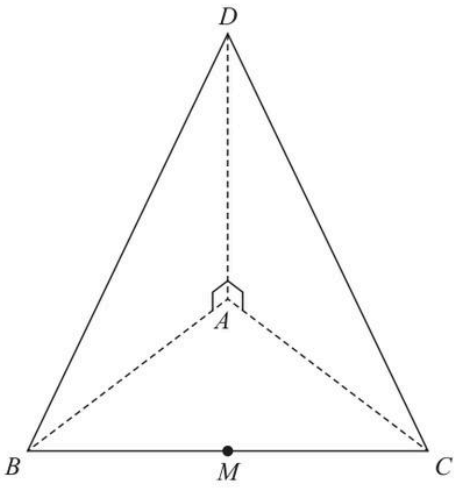


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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10. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

Question 3 continued

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10. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

Question 3 continued

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10. Question 3 · part c · 4 marks

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Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 9 marks)

10. Question 3 · part c · 4 marks · Working and answer

Working and answer

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11. Question 6 · part c · 3 marks

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Further practice · Attempt part c only.

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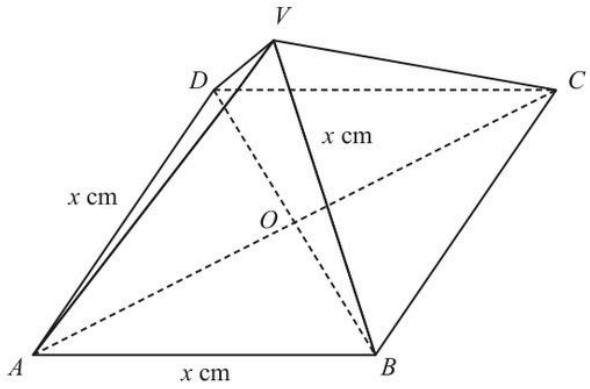


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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11. Question 6 · part c · 3 marks · Working and answer

Working and answer

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